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Gegeven:

$$w = \sin xyz$$

$$x = t$$

$$y = t^2$$

$$z = t^3$$

Gevraagd:

$$\frac{dw}{dt}$$

Volgens de kettingregel geldt:

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt}$$

We berekenen eerst de partiele afgeleiden:

$$\frac{\partial w}{\partial x} = yz \cos xyz$$

$$\frac{\partial w}{\partial y} = xz \cos xyz$$

$$\frac{\partial w}{\partial z} = xy \cos xyz$$

Ook geldt:

$$\frac{dx}{dt} = 1$$

$$\frac{dy}{dt} = 2t$$

$$\frac{dz}{dt} = 3t^2$$

Dus:

$$\frac{dw}{dt} = yz \cos xyz + 2txz \cos xyz + 3t^2 xy \cos xyz$$

$$= (yz + 2txz + 3t^2 xy) \cos xyz$$

Nu vullen we $x = t$, $y = t^2$ en $z = t^3$ in:

$$yz = t^2 t^3 = t^5$$

$$2txz = 2t \cdot t \cdot t^3 = 2t^5$$

$$3t^2xy = 3t^2 \cdot t \cdot t^2 = 3t^5$$

$$xyz = t \cdot t^2 \cdot t^3 = t^6$$

Bijgevolg:

$$\begin{aligned}\frac{dw}{dt} &= (t^5 + 2t^5 + 3t^5) \cos t^6 \\ &= 6t^5 \cos t^6\end{aligned}$$

$$\boxed{\frac{dw}{dt} = 6t^5 \cos t^6}$$

References